

Title: Retail Sales Analytics Using Excel, SQL, Power BI and Python

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Problem Statement:

In a competitive retail market, businesses need to track sales performance, understand customer behavior, and identify growth opportunities. This project aims to build an end-to-end analytics solution that enables insights generation from raw transactional data using a multi-tool approach.

Introduction:

The Retail Sales Analytics project was developed as an end-to-end data analytics solution using Excel, SQL, Python and Power BI. The project started with raw retail datasets containing information about customers, orders, products, stores, staff and inventory. The data was progressively cleaned, structured, analysed and transformed into meaningful business information.

The project was completed in four major phases. Excel was used for initial data cleaning and preparation, MySQL was used for database creation and SQL-based analysis, Python was used for exploratory data analysis and customer segmentation, and Power BI was used to create the final interactive dashboard. The overall workflow helped analyse sales performance, customer purchasing behaviour, product performance, store and staff performance, customer segments and inventory conditions.

Phase 1 – Excel Data Cleaning & Preparation

- ✓ The first phase focused on preparing the raw retail datasets for further analysis. The datasets were reviewed to identify common data quality issues such as duplicate records, missing values, inconsistent formats and incorrect data types.
- ✓ The column names and formats were standardized to maintain consistency across the datasets. Date fields and numerical fields were converted into appropriate formats, and missing or blank values were identified and handled. Duplicate records were also checked, particularly in the transactional data. To prepare the data for analysis, calculated fields were created for sales-related calculations. VLOOKUP/XLOOKUP was used to bring required product information into the transaction data using the product identifier.

A	B	C	D	E	F	G	H	I	J
order_id	item_id	product_id	quantity	list_price	discount	total_price	product_name	category_id	category_name
1346	5	263	2	89.99	0.20	179.78	Strider Classic 12 Balance Bike - 2018	1	Children Bicycles
1379	3	263	2	89.99	0.10	179.88	Strider Classic 12 Balance Bike - 2018	1	Children Bicycles
1425	2	263	1	89.99	0.07	89.92	Strider Classic 12 Balance Bike - 2018	1	Children Bicycles
1439	2	263	1	89.99	0.20	89.79	Strider Classic 12 Balance Bike - 2018	1	Children Bicycles
1462	1	263	1	89.99	0.20	89.79	Strider Classic 12 Balance Bike - 2018	1	Children Bicycles
1520	4	263	2	89.99	0.10	179.88	Strider Classic 12 Balance Bike - 2018	1	Children Bicycles
1573	5	263	2	89.99	0.10	179.88	Strider Classic 12 Balance Bike - 2018	1	Children Bicycles
667	2	84	1	109.99	0.10	109.89	Sun Bicycles Lil Kitt'n - 2017	1	Children Bicycles
690	1	84	1	109.99	0.20	109.79	Sun Bicycles Lil Kitt'n - 2017	1	Children Bicycles
701	1	84	1	109.99	0.05	109.94	Sun Bicycles Lil Kitt'n - 2017	1	Children Bicycles
733	1	84	1	109.99	0.05	109.94	Sun Bicycles Lil Kitt'n - 2017	1	Children Bicycles
734	3	84	2	109.99	0.20	219.78	Sun Bicycles Lil Kitt'n - 2017	1	Children Bicycles
907	1	84	2	109.99	0.07	219.91	Sun Bicycles Lil Kitt'n - 2017	1	Children Bicycles
989	4	84	2	109.99	0.07	219.91	Sun Bicycles Lil Kitt'n - 2017	1	Children Bicycles
1037	5	84	2	109.99	0.05	219.93	Sun Bicycles Lil Kitt'n - 2017	1	Children Bicycles
1137	3	84	2	109.99	0.20	219.78	Sun Bicycles Lil Kitt'n - 2017	1	Children Bicycles

Figure 1: Cleaned Retail Dataset in Excel

- ✓ Pivot Tables were then used to perform initial analysis such as category-level sales analysis. The cleaned datasets were finally prepared in CSV format and used as the input for the SQL phase.

Row Labels	Sum of total_price
Children Bicycles	327803.83
Comfort Bicycles	438452.02
Cruisers Bicycles	1109007.23
Cyclocross Bicycles	799846.1
Electric Bikes	1020214.56
Mountain Bikes	3030651.45
Road Bikes	1852516.12
Grand Total	8578491.31

Figure 2: Category-Level Sales Analysis Using Pivot Table

Key Activities

1. Data cleaning and pre-processing
2. Duplicate and missing-value handling
3. Data type and format correction
4. Derived column creation
5. VLOOKUP/XLOOKUP-based data merging
6. Pivot Table analysis
7. Basic outlier checking
8. Final CSV preparation

Phase 2 – SQL Database Management & Analysis

- ✓ After completing the Excel cleaning process, the prepared datasets were imported into MySQL. The database was structured using separate relational tables for the major entities involved in the retail business.
- ✓ The required tables were created and the cleaned CSV files were imported into the database. The relationships between the tables were considered while performing SQL analysis. Different SQL queries were then developed to answer the business requirements provided for the project.

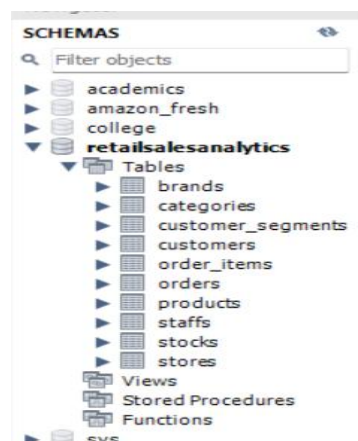
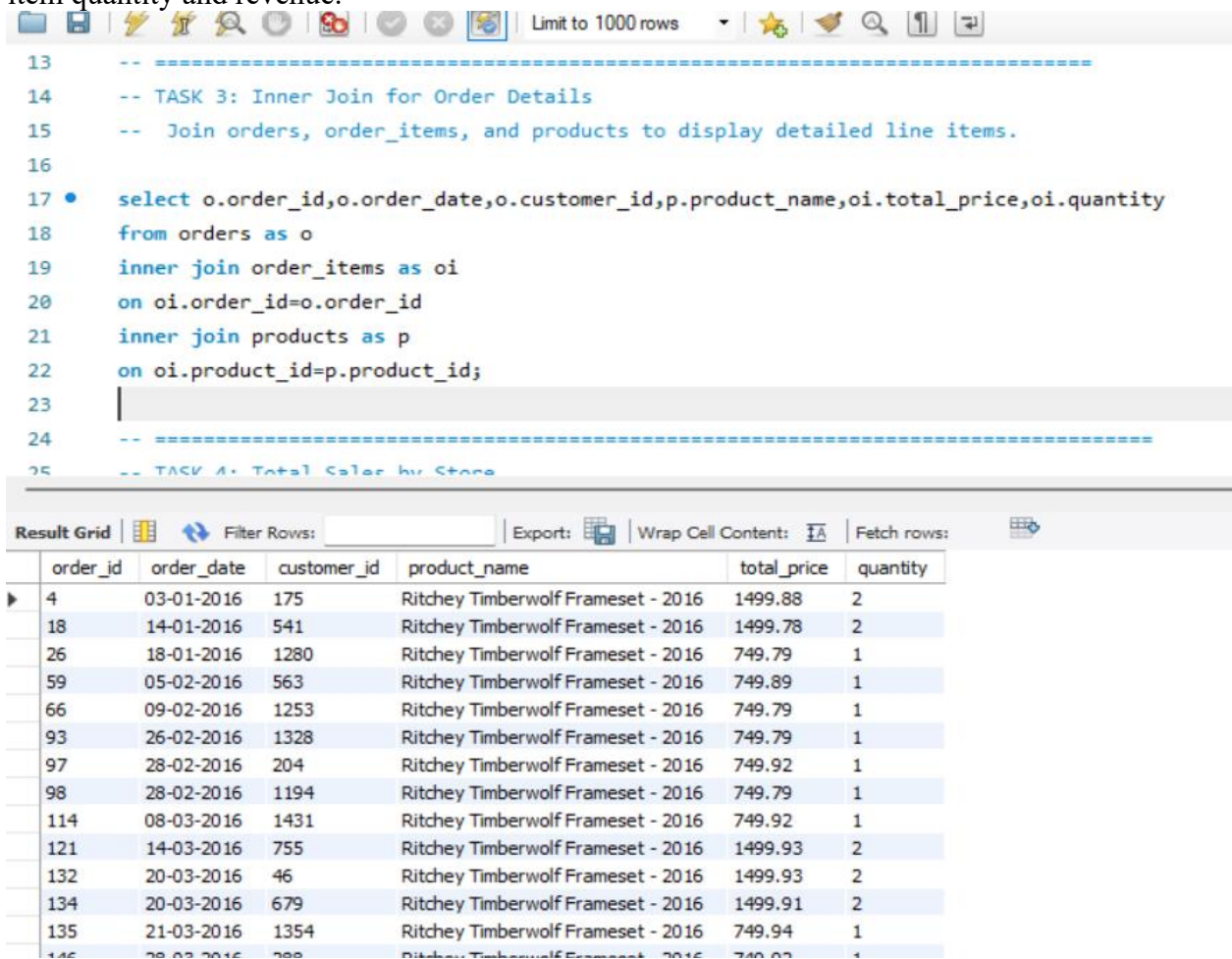


Figure 3: Retail Sales Database and Tables in MySQL

- ✓ Joins were performed between orders, order_items and products to generate detailed order information. Sales revenue was analysed at store level, and the Top 5 Selling Products were identified. A customer purchase summary was also created by calculating customer-level order count, item quantity and revenue.



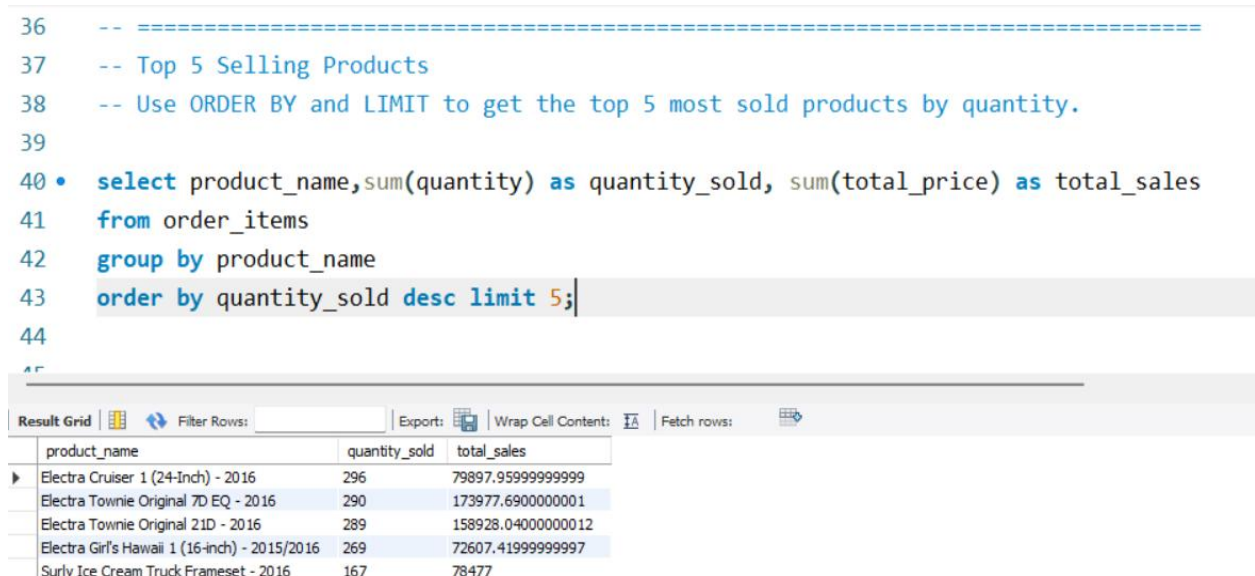
```

13  -- =====
14  -- TASK 3: Inner Join for Order Details
15  -- Join orders, order_items, and products to display detailed line items.
16
17  • select o.order_id,o.order_date,o.customer_id,p.product_name,oi.total_price,oi.quantity
18  from orders as o
19  inner join order_items as oi
20  on oi.order_id=o.order_id
21  inner join products as p
22  on oi.product_id=p.product_id;
23
24  -- =====
25  -- TASK 4: Total Sales by Store

```

	order_id	order_date	customer_id	product_name	total_price	quantity
▶	4	03-01-2016	175	Ritchey Timberwolf Frameset - 2016	1499.88	2
	18	14-01-2016	541	Ritchey Timberwolf Frameset - 2016	1499.78	2
	26	18-01-2016	1280	Ritchey Timberwolf Frameset - 2016	749.79	1
	59	05-02-2016	563	Ritchey Timberwolf Frameset - 2016	749.89	1
	66	09-02-2016	1253	Ritchey Timberwolf Frameset - 2016	749.79	1
	93	26-02-2016	1328	Ritchey Timberwolf Frameset - 2016	749.79	1
	97	28-02-2016	204	Ritchey Timberwolf Frameset - 2016	749.92	1
	98	28-02-2016	1194	Ritchey Timberwolf Frameset - 2016	749.79	1
	114	08-03-2016	1431	Ritchey Timberwolf Frameset - 2016	749.92	1
	121	14-03-2016	755	Ritchey Timberwolf Frameset - 2016	1499.93	2
	132	20-03-2016	46	Ritchey Timberwolf Frameset - 2016	1499.93	2
	134	20-03-2016	679	Ritchey Timberwolf Frameset - 2016	1499.91	2
	135	21-03-2016	1354	Ritchey Timberwolf Frameset - 2016	749.94	1
	146	28-03-2016	288	Ritchey Timberwolf Frameset - 2016	749.92	1

Figure 4: Order Details Generated Using SQL Joins



```

36  -- =====
37  -- Top 5 Selling Products
38  -- Use ORDER BY and LIMIT to get the top 5 most sold products by quantity.
39
40  • select product_name,sum(quantity) as quantity_sold, sum(total_price) as total_sales
41  from order_items
42  group by product_name
43  order by quantity_sold desc limit 5;
44
45

```

	product_name	quantity_sold	total_sales
▶	Electra Cruiser 1 (24-Inch) - 2016	296	79897.95999999999
	Electra Townie Original 7D EQ - 2016	290	173977.69000000001
	Electra Townie Original 21D - 2016	289	158928.04000000012
	Electra Girl's Hawaii 1 (16-inch) - 2015/2016	269	72607.41999999997
	Surly Ice Cream Truck Frameset - 2016	167	78477

Figure 5: Top 5 Selling Products Based on Revenue

- ✓ Customer spending was further classified into different spending categories using SQL conditions. Staff performance was analysed based on the revenue generated from their handled orders. A stock alert query was also created to identify products with low inventory levels.

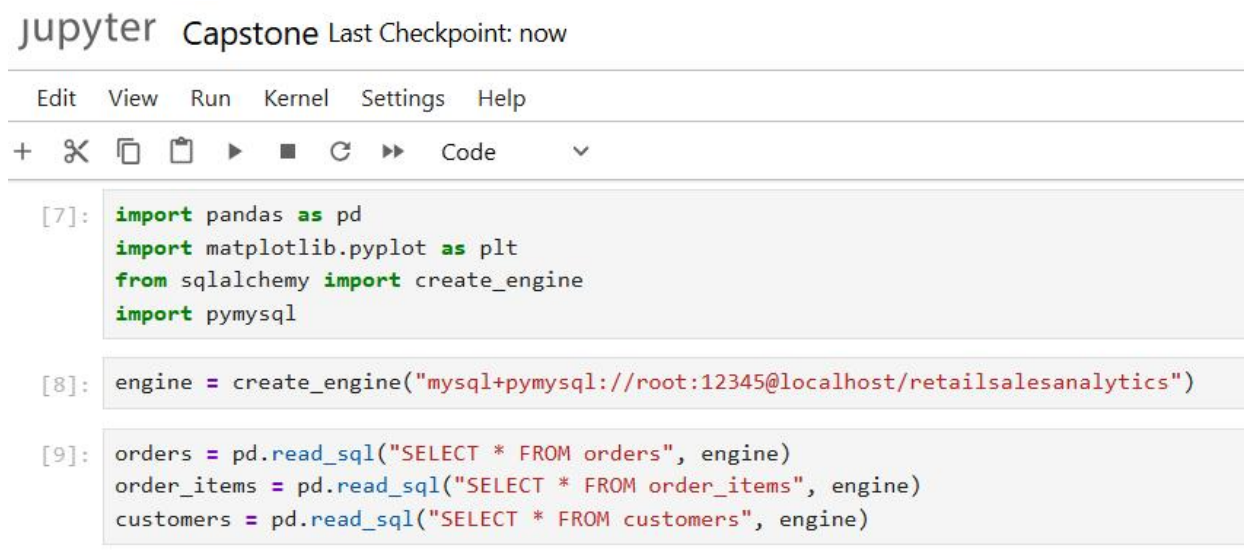
- ✓ Finally, a separate customer_segments table was created to store the customer segmentation results that would later be generated using Python.

Key Activities

1. Database and table creation
2. CSV data import
3. SQL joins and aggregations
4. Store-wise sales analysis
5. Top 5 product analysis
6. Customer purchase summary
7. Customer spending classification
8. Staff performance analysis
9. Low-stock analysis
10. customer_segments table creation

Phase 3 – Python EDA & Customer Segmentation

- ✓ The third phase focused on analysing customer behaviour using Python. The required data was retrieved from the MySQL database using Pandas, with the main analysis based on the orders, order_items and customers tables.



The screenshot shows a Jupyter Notebook window titled 'jupyter Capstone' with the status 'Last Checkpoint: now'. The interface includes a menu bar (Edit, View, Run, Kernel, Settings, Help) and a toolbar with icons for file operations and execution. Three code cells are visible:

```
[7]: import pandas as pd
import matplotlib.pyplot as plt
from sqlalchemy import create_engine
import pymysql

[8]: engine = create_engine("mysql+pymysql://root:12345@localhost/retailsalesanalytics")

[9]: orders = pd.read_sql("SELECT * FROM orders", engine)
order_items = pd.read_sql("SELECT * FROM order_items", engine)
customers = pd.read_sql("SELECT * FROM customers", engine)
```

Figure 6: Retail Data Imported into Python Using Pandas

- ✓ The imported datasets were first explored to understand their structure and contents. Pandas was used for data manipulation and analysis, while Matplotlib and Seaborn were used to visualize important patterns in the data.

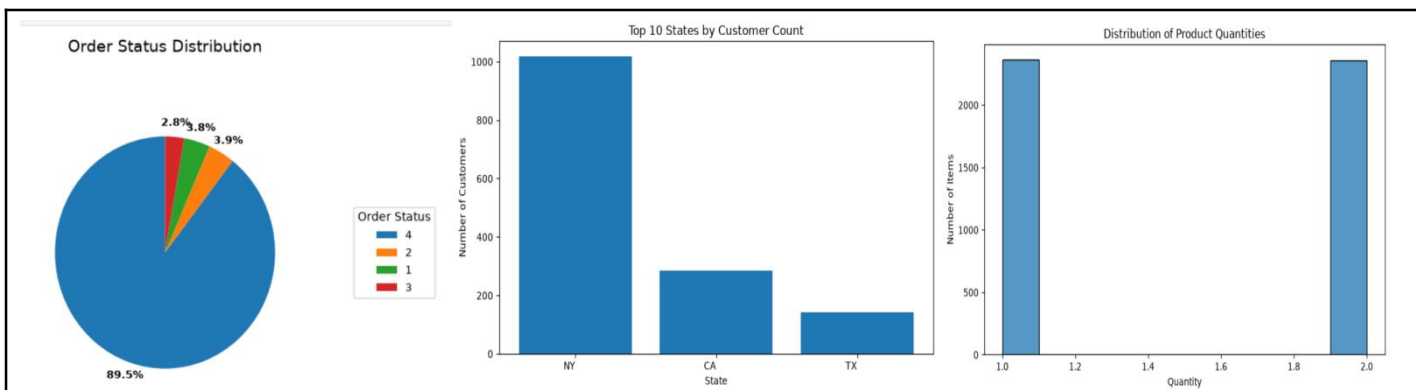


Figure 7: Exploratory Data Analysis Visualizations Using Matplotlib and Seaborn

✓ The major focus of this phase was customer segmentation using RFM analysis. Three important customer behaviour metrics were calculated:

1. Recency – how recently a customer made a purchase.
2. Frequency – how frequently a customer placed orders.
3. Monetary – how much revenue a customer generated.

```
rfm_data.columns = ['customer_id', 'Recency', 'Frequency', 'Monetary']
```

```
rfm_data.head()
```

	customer_id	Recency	Frequency	Monetary
0	1	41	3	30644.78
1	2	264	3	21652.87
2	3	69	3	26248.66
3	4	255	3	24197.81
4	5	256	3	19442.01

Figure 8: RFM Analysis for Customer Behaviour

✓ These RFM features were used to understand differences in customer purchasing behaviour and group customers into meaningful segments. The resulting customer segment information was then exported back to the MySQL database and stored in the customer_segments table.

```
38]: rfm_data['Segment'].value_counts()
```

```
38]: Segment
Regular    721
Loyal      362
At Risk    362
Name: count, dtype: int64
```

Figure 9: Customer Segmentation Results

✓ This created a connection between the Python analysis and the final Power BI dashboard.

Phase 4 – Power BI Visualization & Dashboarding

✓ The final phase focused on converting the analysed data into an interactive Power BI dashboard. The required tables were connected from MySQL and relationships were established between the tables

based on their keys. The dashboard was designed to provide a consolidated view of the retail business through KPI cards, charts, tables, maps and interactive filters.

Sales Overview

✓ The Sales Overview section was developed to monitor the overall sales performance of the business. It includes KPI cards for important measures such as:

1. Total Revenue
2. Total Orders
3. Average Order Value
4. Total Customers

✓ Sales trends and monthly performance were also visualized to understand changes in revenue over time. A Top Products visual was included to identify products contributing significantly to overall revenue.

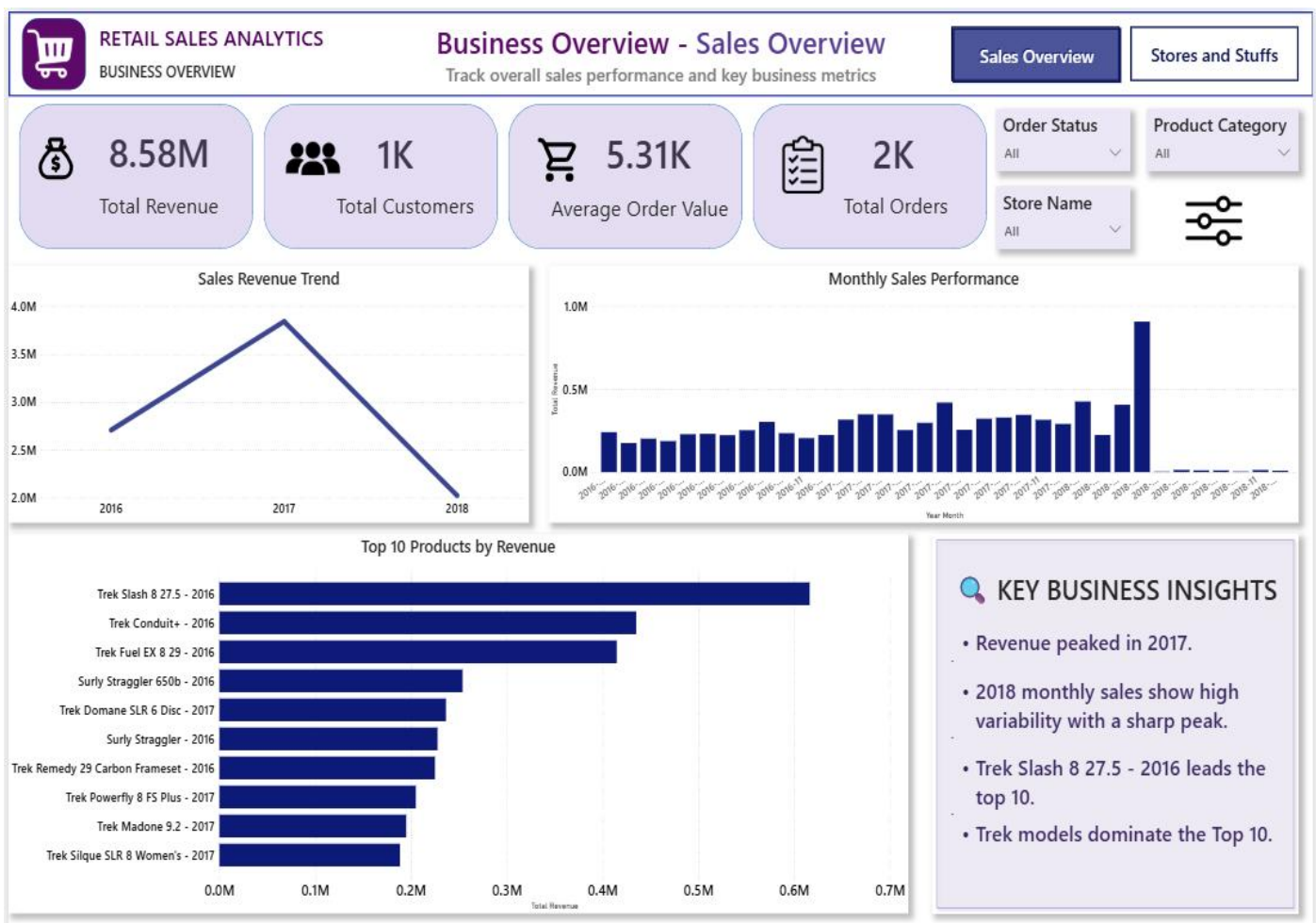


Figure 10: Sales Overview Dashboard

Stores & Staffs Analysis

✓ A separate Stores & Staffs section was created to analyse operational performance. Store-level sales were visualized geographically using a map, while staff-level revenue was analysed to compare individual performance. A Low Stock Alert section was also included to identify products with lower inventory levels and support inventory monitoring.

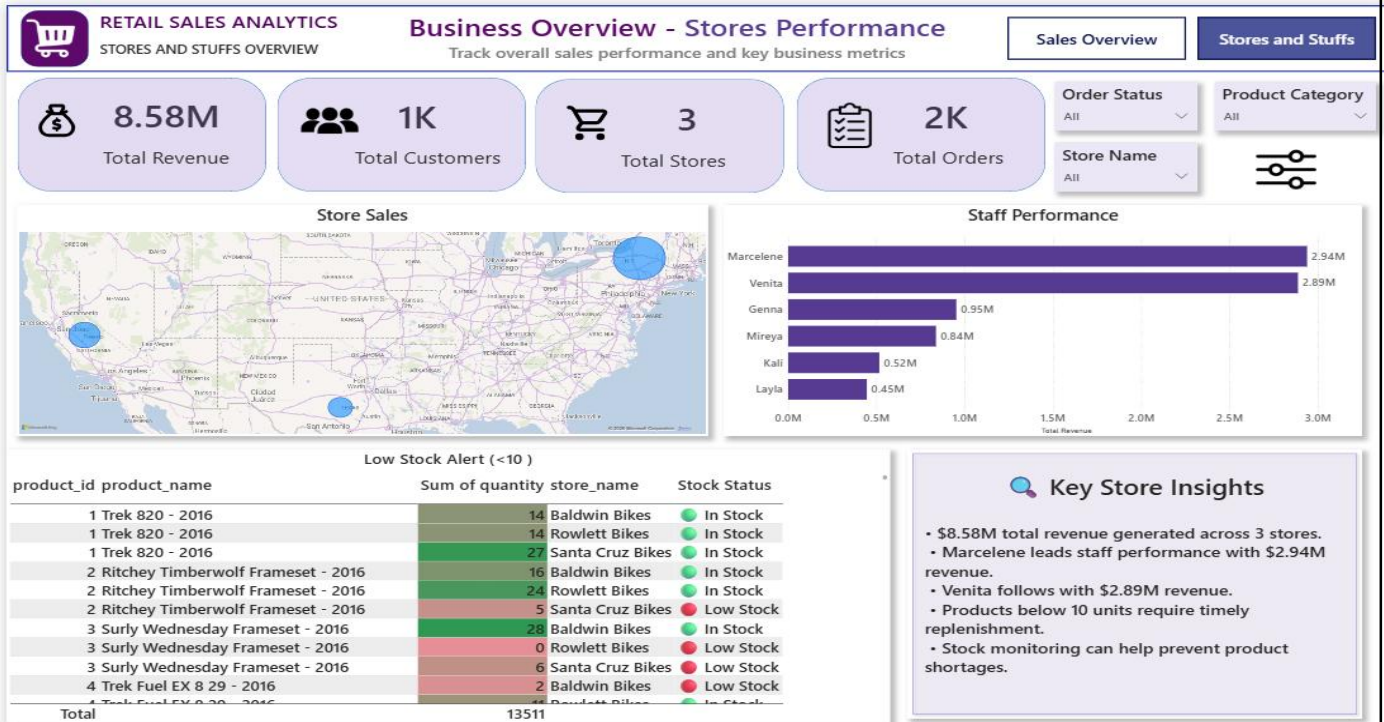


Figure 11: Stores and Staff Performance Dashboard

Customer Analysis

- ✓ The Customer Analysis section focused on understanding purchasing behaviour and customer segments. A stacked bar chart was used to analyse purchase patterns across cities. Customer behaviour was further explored using the customer analysis data.
- ✓ The customer segmentation results generated through Python were incorporated into Power BI. A donut chart was used to show the distribution of customer segments, while a bar chart was used to compare revenue across customer segments. A Segment slicer was also added so that the report could be interactively filtered based on customer segment.

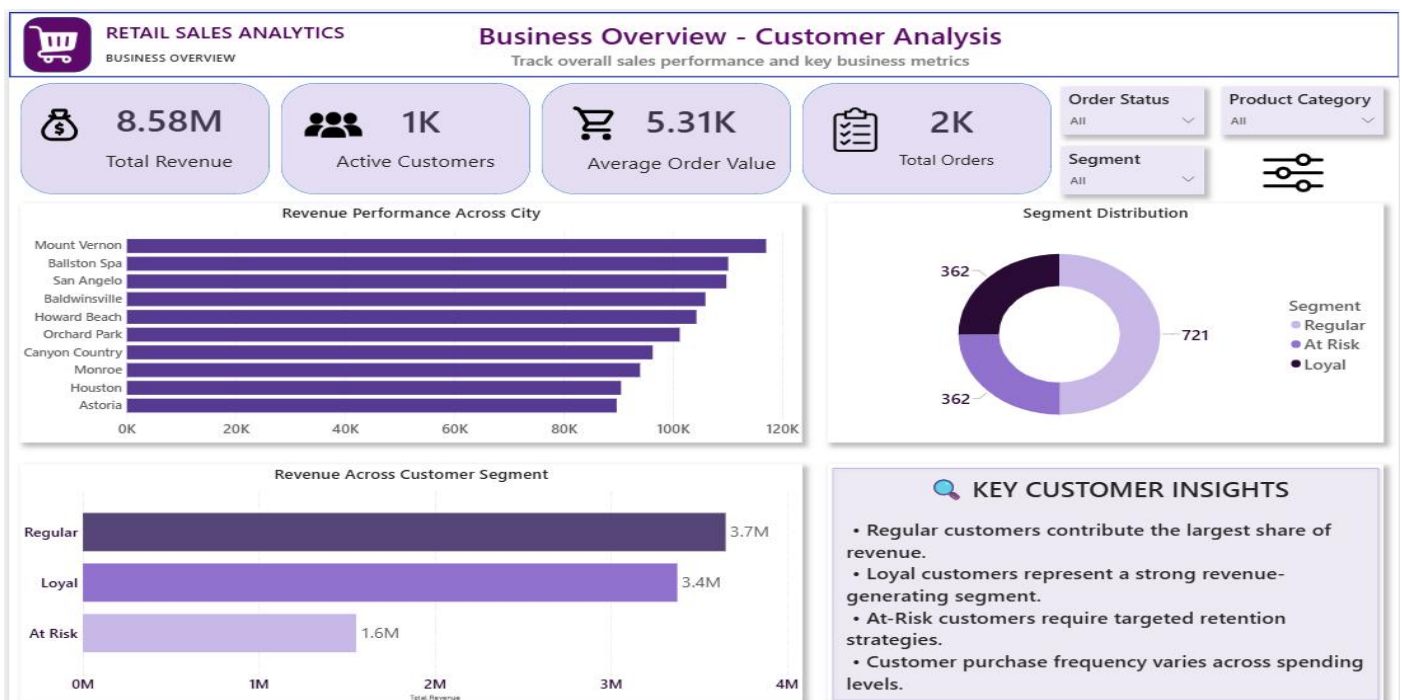


Figure 12: Customer Analysis Dashboard

Key Insights

1. Revenue peaked at \$3.8M in 2017, followed by a decline in 2018.
2. Trek models dominate the Top 10 products, with Trek Slash 8 27.5 - 2016 generating approximately \$0.62M.
3. Marcelene and Venita are the top-performing staff, contributing \$2.94M and \$2.89M respectively.
4. Regular and Loyal customers generate the highest segment revenue at \$3.7M and \$3.4M.
5. The At-Risk segment contributes \$1.6M, highlighting an opportunity for targeted customer retention.
6. Mount Vernon records the highest revenue among the displayed cities at approximately \$120K.
7. Multiple products have stock below 10 units, including zero-stock items, requiring timely replenishment.

Conclusion

The Retail Sales Analytics project successfully implemented a complete analytics workflow starting from raw data preparation and ending with an interactive Power BI dashboard. Excel was used to clean and prepare the datasets, MySQL was used to store and analyse the structured data, Python was used for exploratory analysis and RFM-based customer segmentation, and Power BI was used to present the results through interactive dashboards.

The project provided practical experience in data cleaning, SQL analysis, Python-based data analysis, customer segmentation, visualization and business intelligence, while bringing the outputs from all four phases together into a single retail analytics solution.

Tools & Technologies Used

Microsoft Excel | MySQL | Python | Pandas | NumPy | Matplotlib | Seaborn | Power BI